BCSE308P COMPUTER NETWORKS LAB · INNOVATIVE EVALUATION PROJECT

Contender-Estimation Adaptive Backoff

Teaching Wi-Fi stations how long to wait

A comparative study of nine CSMA/CA backoff strategies in IEEE 802.11, and a new scheme that sets the contention window from the number of stations it overhears

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Crowded Wi-Fi loses a quarter of its capacity

Standard 802.11 binary exponential backoff (BEB), 50 saturated stations, 802.11g 54 Mb/s

−25 %

throughput lost against the best backoff at N = 50 (22.3 vs 29.8 Mb/s)

0.59

probability that a transmission attempt collides

304 ms

99th-percentile access delay: one frame in a hundred waits a third of a second

Why it happens

- After every success BEB resets the window to 16 slots, regardless of how many stations compete
- The next frame collides, doubles, collides again: contention is re-learned for every frame
- The winner keeps the smallest window and tends to win again (capture effect)

Why it matters

- Lecture halls, hostels and IoT deployments put dozens of active stations on one access point
- Short-term fairness at N = 50 is only 0.40 (Jain index; 1.0 = perfectly fair)
- Tail delay, not the mean, is what breaks calls, games and video

Existing waiting strategies either forget or assume

Baseline: IEEE 802.11 DCF with binary exponential backoff

Baseline: BEB

1. Draw backoff b ~ U[0, W−1]
2. Count down idle 9 µs slots, freeze while busy
3. Transmit at b = 0; wait for the ACK
4. No ACK → W ← min(2W, 1024), retry ≤ 7
5. ACK → W ← 16 (forget everything)

History-based rules

Linear · Fibonacci · Quadratic · MILD · EIED · Constant

- Change W by a fixed rule after each outcome
- Never learn how many stations compete
- Linear/Fibonacci collide even more than BEB; MILD can lock at CWmax

Estimation-based

Calì et al. 2000 · Bianchi & Tinnirello 2003 · Idle Sense 2005

- Aim for the optimal window, but model the other stations
- Model inversion is fragile: real counters are correlated
- Idle Sense next to legacy stations gets 0.9 Mb/s vs 27 Mb/s

CEAB counts who is talking, then waits

Contender-Estimation Adaptive Backoff: four steps, run independently by every station

1

Overhear

Log the sender address (TA) of the last M = 200 decoded frames

information every 802.11 header already carries

2

Estimate

$n_{eff} = M(M-1) / \Sigma c_j(c_j-1)$

inverse Simpson index: equals n for n equal senders

3

Correct for load

$\rho = \min(1, I^* / I_{obs})$; $n_c = \rho \cdot n_{eff}$

long idle gaps → fewer backlogged stations

4

Set the window

$W^* = 2 / \tau^*(n_c) - 1$

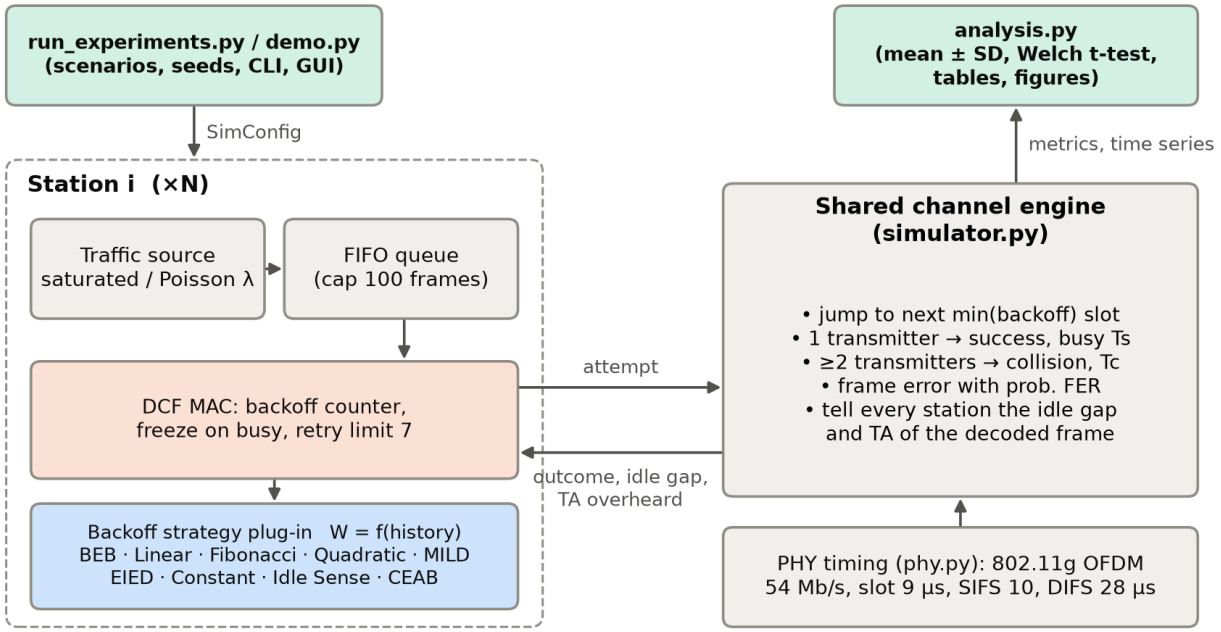
τ^* = throughput-optimal attempt probability (Bianchi)

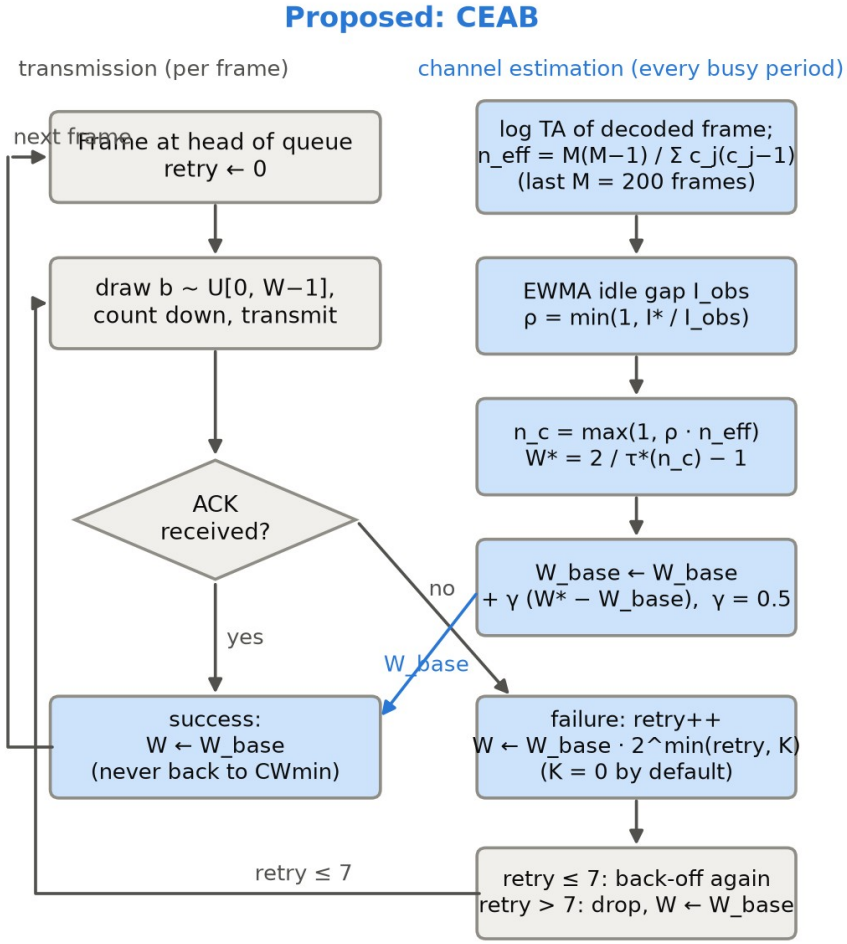
What exactly differs from the baseline

- **After success:** W stays at the optimum, never back to 16
- **After collision:** W unchanged (K = 0), not doubled
- **Estimator:** model-free; valid whatever the others run
- **Same input for all:** all stations hold the same window → fairness

Architecture and algorithm

Event-driven 802.11 DCF simulator with pluggable backoff strategies





Engine: jumps straight to the next slot in which a backoff counter hits zero, so 2170 ten-second runs take about 6 minutes on 16 cores

Implementation setup

Python 3.14 · NumPy · SciPy · Matplotlib · 9 unit tests incl. Bianchi validation

Parameter	Value
PHY	802.11g ERP-OFDM, 54 Mb/s data, 24 Mb/s ACK
Slot / SIFS / DIFS	9 / 10 / 28 µs
Payload	1500 B (256–2304 B sweep)
Ts / Tc (basic)	314 / 323 µs
CWmin / CWmax / retries	15 / 1023 / 7
Stations N	2 – 100
Traffic	saturated, or Poisson 2 – 40 Mb/s
Runs	10 seeds × 10 s, 1 s warm-up
CEAB	M = 200, γ = 0.5, β = 0.05, K = 0

Code structure

`phy.py`	frame timing, Ts and Tc
`analytical.py`	Bianchi model, τ*(n) table
`strategies.py`	9 strategies incl. CEAB
`simulator.py`	event-driven DCF engine
`experiments.py`	2170 parallel runs
`analysis.py`	mean ± SD, t-tests, plots
`demo.py`	live CLI, trace, GUI

Scenarios and metrics

2170 independent simulation runs, every point = mean ± SD of 10 seeds

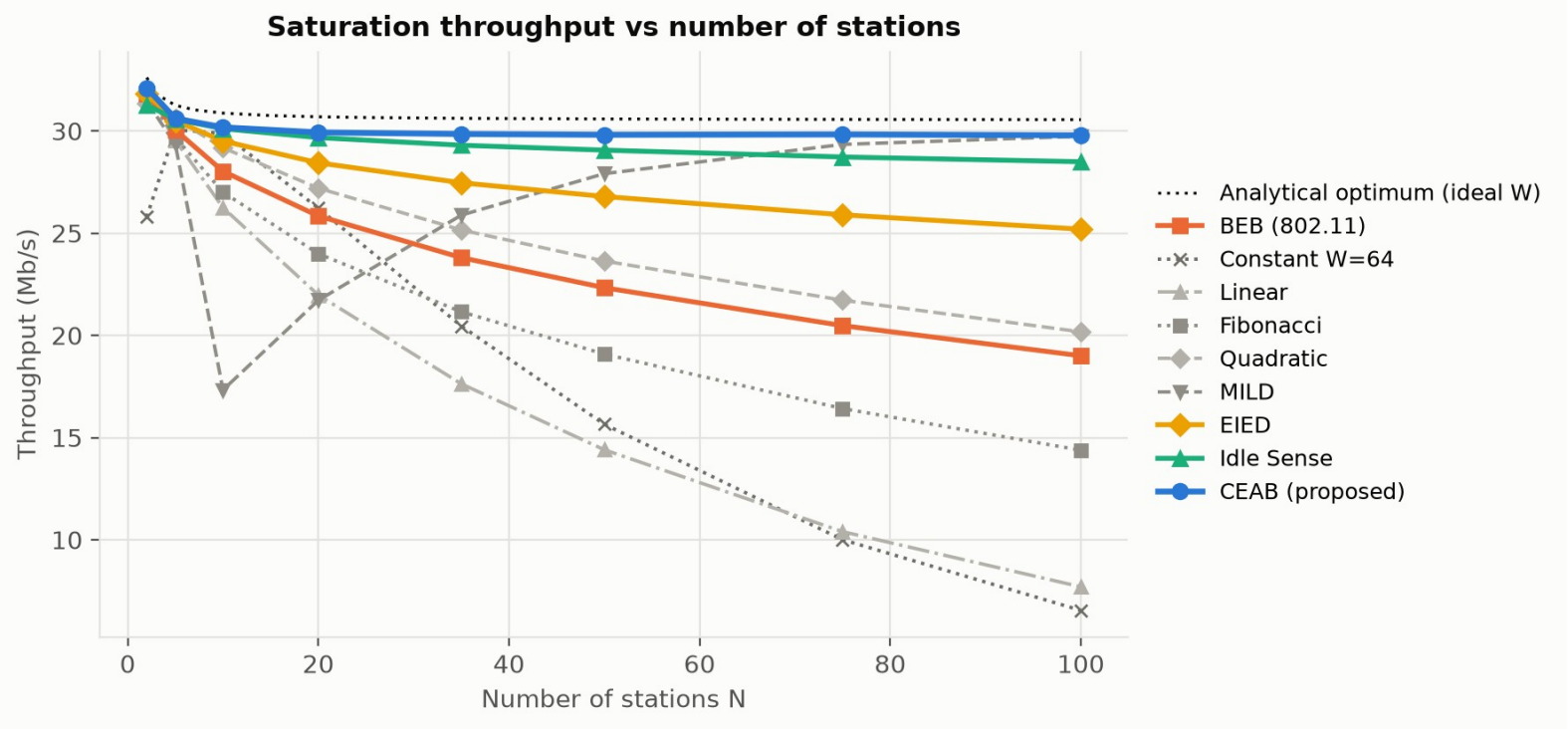
1	**S1 · S2 · S3**	saturated, N = 5 / 20 / 50 (low / medium / high)
2	**Station sweep**	N = 2 … 100, all nine strategies
3	**Offered load**	Poisson traffic, 30 stations, 2 – 40 Mb/s
4	**Robustness**	frame errors 0 – 30 %, RTS/CTS, payload 256 – 2304 B
5	**Churn**	5 → 50 → 15 stations during a run
6	**Coexistence**	10 legacy BEB + 10 new-scheme stations
7	**Ablation**	CEAB escalation cap K and window M

Metrics and why

- **Throughput** — capacity; what the optimal window maximises
- **Collision probability** — direct sign of a window that is too small
- **99th-pct access delay** — the tail is what real-time traffic feels
- **Short-term Jain fairness** — 100 ms windows expose the capture effect
- **Drop rate, end-to-end delay** — losses and queueing seen by upper layers
- **Welch t-test** — is the difference real, not noise?

CEAB stays near capacity as the network grows

Saturation throughput, all nine strategies, N = 2 … 100 (10 runs per point)



+33.5 %

throughput at N = 50 (22.3 → 29.8 Mb/s), +57 % at N = 100

−79 %

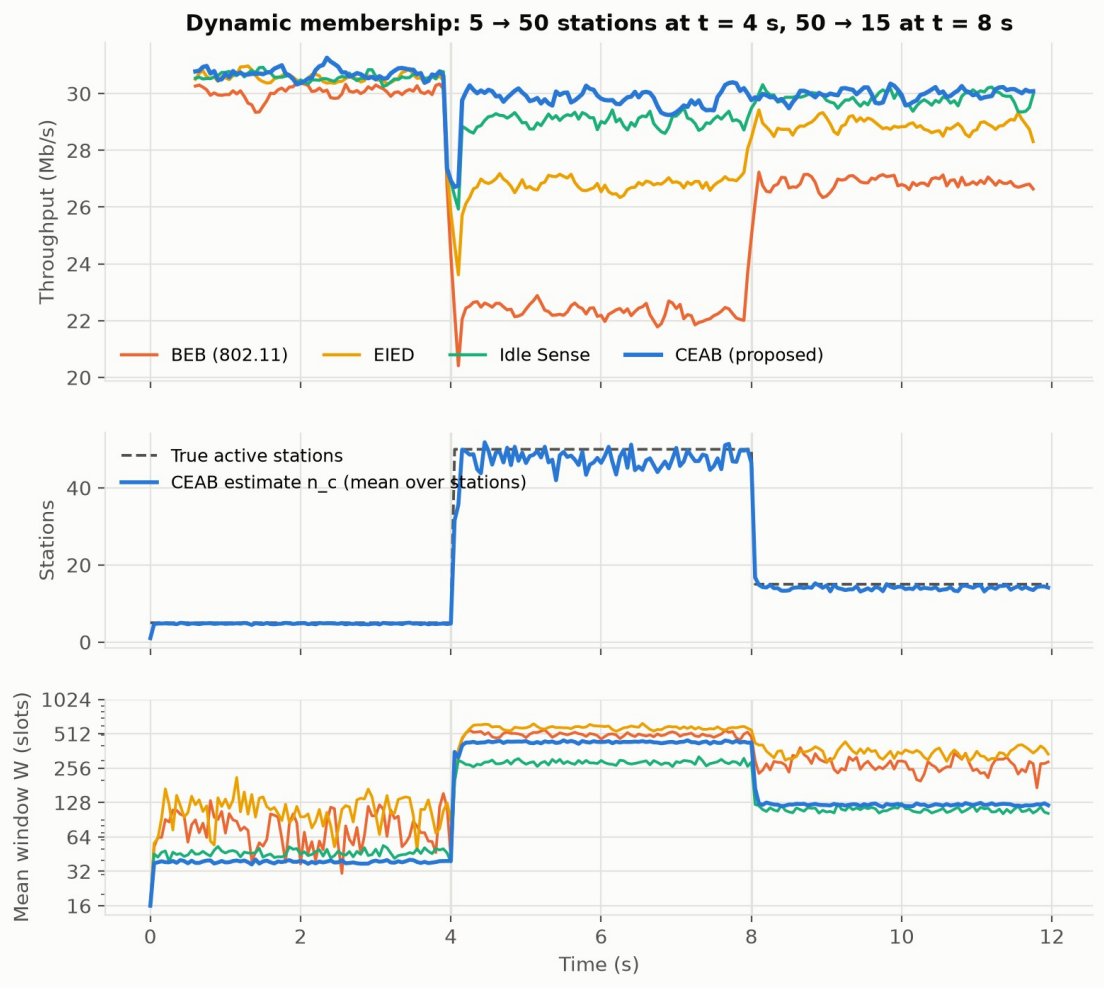
99th-pct delay at N = 50 (304 → 65 ms)

0.40 → 0.91

short-term fairness at N = 50; collisions 0.59 → 0.20

Where it helps, and where it does not

Dynamic membership trace (left) and critical evaluation (right)



Strengths

- Tracks 5 → 50 → 15 stations within ~0.1 s
- Carries 29.9 Mb/s of Poisson traffic vs 24.4 for BEB; light-load delay unchanged
- Safe to adopt. Mixed with old standard stations, a CEAB device still gets more than it would in an all-standard network(Idle Sense starves)

Limitations and failure cases

- RTS/CTS, N = 5: no gain (-0.3 %) — collisions are cheap
- At 30% frame errors gains vanish
- Medium Poisson load: mean delay 0.94 vs 0.77 ms
- Hidden terminals not modelled; old stations still keep a 2.6 to 1 share against CEAB stations.

Conclusion, future work and references

Conclusion

- Waiting well = remembering how crowded the channel is
- CEAB: +16 % / +33 % throughput at N = 20 / 50, −79 % tail delay, fairness 0.91
- Within 2.5 % of the theoretical limit for 5 – 100 stations
- Uses only information every Wi-Fi receiver already decodes, the sender address in each frame.

Future work

- Hidden terminals and multi-cell; airtime-aware estimate for multi-rate
- Multi-speed networks, counting airtime instead of frames
- Porting it to real open-source firmware for a hardware test
- Large IoT networks with hundreds of devices, where the gains should be even larger

Key references

[1] IEEE Std 802.11-2020

[2] G. Bianchi, IEEE JSAC 18(3), 2000

[3] F. Calì et al., IEEE/ACM ToN 8(6), 2000

[4] V. Bharghavan et al., MACAW, SIGCOMM 1994

[5] N.-O. Song et al., EIED, IEEE VTC 2003

[6] M. Heusse et al., Idle Sense, SIGCOMM 2005

[7] G. Bianchi, I. Tinnirello, INFOCOM 2003

[16] E. H. Simpson, Nature 163, 1949

Full IEEE-style list: project report, Ch. 7